

Introduction to Crystal Maths

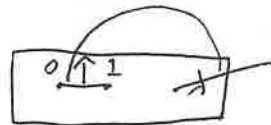
Ben Church

$$\text{Spec} \left(k[\lambda][x, y] / (y^2 - x(x-1)(x-\lambda)) \right) \rightarrow \text{Spec} \left(k[\lambda][\lambda^{-1}(1-\lambda)^{-1}] \right)$$

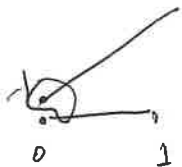
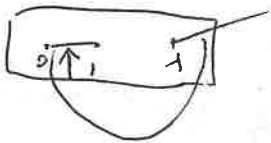
$\underline{k = \mathbb{C}}$ $\pi_1(\lambda) = 2 \int_0^1 \omega$, $\pi_2(\lambda) = 2 \int_1^\lambda \omega$ meromorphic function of λ .

$$\omega = \frac{dx}{y}$$

$$\pi(\lambda) = \pi_1(\lambda) + \pi_2(\lambda) = 2 \int_0^\lambda \omega$$



$$y = \sqrt{x(x-1)(x-\lambda)}$$



$$\lambda \rightarrow 0, \quad \pi(\lambda) \rightarrow \left(\text{res}, \frac{dx}{y} \right) 2\pi i = 2\pi$$

holomorphic at $\lambda = 0$.

$\omega, \frac{\partial \omega}{\partial \lambda}, \frac{\partial^2 \omega}{\partial \lambda^2}$ up to $d(f)$ and $d\lambda$.

$$\frac{\partial \omega}{\partial \lambda} = \frac{dx}{2 \sqrt{x(x-1)(x-\lambda)^3}}, \quad \frac{\partial^2 \omega}{\partial \lambda^2} = \frac{3 dx}{4 \sqrt{x(x-1)(x-\lambda)^5}}$$

$$d\left(\frac{y}{(x-\lambda)^2}\right) = \frac{1}{4} \omega + (2\lambda-1) \frac{\partial \omega}{\partial \lambda} + \lambda(\lambda-1) \frac{\partial^2 \omega}{\partial \lambda^2} + () d\lambda$$

$\oint_\gamma$ gives $0 = \frac{1}{4} \pi i + (2\lambda-1) \frac{\partial \pi i}{\partial \lambda} + \lambda(\lambda-1) \frac{\partial^2 \pi i}{\partial \lambda^2}$ Picard-Fuchs equation

near $\lambda=0$, $\sigma_1(\lambda), \sigma_2(\lambda) + (\log \lambda) \sigma_1(\lambda)$

σ_i holomorphic near $\lambda=0$

$$\begin{aligned} \sigma_1 &= \sum_{n \geq 0} a_n \lambda^n, \quad \sum_{n \geq 0} \left[\lambda(\lambda-1)(n+1)(n+2) a_{n+2} + (2\lambda-1)(n+1) a_{n+1} + \frac{1}{4} a_n \right] \lambda^n = 0 \\ &= \sum \left[(n+\frac{1}{2})^2 a_n - (n+1)^2 a_{n+1} \right] \lambda^n = 0 \end{aligned}$$

$$a_{n+1} = \left(\frac{n+1/2}{n+1} \right)^2 a_n, \quad a_0 = 1$$

$$a_n = \left(\frac{-1/2}{n} \right)^2, \quad \left(\frac{-1/2}{n} \right) = \frac{(-1) \cdots (-2n+1)}{2^n n!}$$

$$\sigma_1(\lambda) = \sum_{n \geq 0} \left(\frac{-1/2}{n} \right)^2 \lambda^n = F(1/2, 1/2, 1, -\lambda) \quad \begin{array}{l} \text{holomorphic at } \lambda=0 \\ \text{and } p\text{-adic analytic } \boxed{p \neq 2} \end{array}$$

$$k = \mathbb{F}_p, \quad p > 2, \quad \lambda \in \mathbb{F}_p, \quad \# C_\lambda \quad y^2 = f(x), \quad f(x) \text{ sq.}$$

$$\Leftrightarrow f(x)^{\frac{p-1}{2}} \equiv 1 \pmod{p}$$

$$\# C_\lambda = \sum_{x \in \mathbb{F}_p} \left(1 + f(x)^{\frac{p-1}{2}} \right) \equiv -a_{p-1} \pmod{p}$$

$$f(x)^{\frac{p-1}{2}} = \sum_{n \geq 0} a_n x^n, \quad f(x)^{\frac{p-1}{2}} = x^{\frac{p-1}{2}} \underbrace{(x-1)^{\frac{p-1}{2}} (x-\lambda)^{\frac{p-1}{2}}}_{x^{\frac{p-1}{2}}}$$

$$\left[\sum_{k=0}^{\frac{p-1}{2}} \binom{\frac{p-1}{2}}{k} (-1)^k x^{\frac{p-1}{2}-k} \right] \left[\sum_{l=0}^{\frac{p-1}{2}} \binom{\frac{p-1}{2}}{l} (-\lambda)^l x^{\frac{p-1}{2}-l} \right]$$

$$x^{\frac{p-1}{2}} \text{ coeff} = (-1)^{\frac{p-1}{2}} \sum_{k+l=\frac{p-1}{2}} \binom{\frac{p-1}{2}}{k} \binom{\frac{p-1}{2}}{l} \lambda^l$$

$$= (-1)^{\frac{p-1}{2}} \sum_{k=0}^{\frac{p-1}{2}} \binom{\frac{p-1}{2}}{k}^2 \lambda^k$$

$$= (-1)^{\frac{p-1}{2}} \sum_{k=0}^{\frac{p-1}{2}} \left(\frac{-1/2}{k} \right)^2 \lambda^k \pmod{p}$$

§ Motivation $X/\overline{\mathbb{F}_p} \rightsquigarrow H_{\text{ét}}^1(X, \mathbb{Z}_\ell) \quad \mathbb{Z}_\ell\text{-module.}$

detect. "l-torsion" in cohomology of X , $\ell \neq p$

Ex. $X/\mathbb{C}$ Enriques surface. $H_{\text{sing}}^2(X, \mathbb{Z})_{\text{tors}} = \mathbb{Z}/2$

$$H_{\text{ét}}^2(X/\overline{\mathbb{F}_\ell}, \mathbb{Z}_\ell)_{\text{tors}} = \mathbb{Z}/2$$

$$X/\overline{\mathbb{F}_\ell}, H_{\text{ét}}^2(X, \mathbb{Z}_\ell)_{\text{tors}} = 0, \text{ but } H_{\text{dR}}^1(X) = k = \overline{\mathbb{F}_2}.$$

$E/\overline{\mathbb{F}_p}$ super singular.

Lemma $R = \text{End}(E) \otimes \mathbb{Q}$ is the unique quaternion alg, $\mathbb{Q}_{p,\infty}$ non split at exactly p, ∞ .

Proof. R is a division ring, $\deg: R \rightarrow \mathbb{Q}$ mult.

$$R \hookrightarrow T_\ell E \text{ faithfully } R \hookrightarrow \text{GL}_2(\mathbb{Q}_\ell) = M_2(\mathbb{Q}_\ell)$$

$$\dim R \leq 4$$

$$\text{if } \dim_{\mathbb{Q}} R > 2 \Rightarrow R \otimes \mathbb{Q}_\ell = M_2(\mathbb{Q}_\ell) \text{ for all } \ell \neq p.$$

but R not $M_2(\mathbb{Q})$ b/c no zero divisors.

$$R \not\hookrightarrow M_2(\mathbb{Q}_p)$$

$H^i(-, k)$ is a cohomology theory $\text{Sm}_{\overline{\mathbb{F}_p}} \rightarrow \text{Vect}_k$

— Künneth

— has wrong Betti # for curves

$\text{End}(E) \rightarrow M_2(k)$, thus k must split $\mathbb{Q}_{p,\infty}$.

defined only
over $\mathbb{F}_{p^2}$ $\text{Frac}(W(\mathbb{F}_{p^2})) = \mathbb{Q}_{p^2}$

$$\rightsquigarrow H^i(-, W(k)).$$

§ Lifts

k perfect char = p ($\mathbb{F}_q$) $f: X \rightarrow Y$ smooth var. / k

$$\begin{array}{ccc} X & \hookrightarrow & \tilde{X} \\ \downarrow & & \downarrow \\ \mathrm{Spec} k & \hookrightarrow & \mathrm{Spec} W \end{array}$$

$H_{dR}^i(\tilde{X}/W)$ valued in W -modules.

is $H_{\mathrm{crys}}^i(X/W)$ when defined up to isom.

depending on $\tilde{X}$:

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ \tilde{X} & \xrightarrow{\tilde{f}} & \tilde{Y} \end{array}$$

$$\text{then } \tilde{f}^*: H_{dR}^i(\tilde{Y}/W) \rightarrow H_{dR}^i(\tilde{X}/W)$$

only depends on f , not on choice of lift.
don't even need $\tilde{f}$.

Suppose $\tilde{f}_1, \tilde{f}_2$ over $W_2(k)$. $\tilde{f}_1 - \tilde{f}_2 = p\zeta$, $\zeta \in H^0(f^*T_Y)$

$$H_{dR}^i \text{ computed via } C^{p,q} = \bigoplus_{i_0, \dots, i_p} H^0(\tilde{U}_{i_0, \dots, i_p}, \Omega_{\tilde{X}/W}^q)$$

$$(\tilde{f}_1^* - \tilde{f}_2^*)\omega = pL_{\zeta}^* \omega = p(L_{\zeta} d t^* \omega + d L_{\zeta} t^* \omega)$$

$$d^{\mathrm{tot}} = d + \delta_{\mathrm{Frob}} (-1)^q, \quad [L_{\zeta}, \delta] = 0.$$

(
in fact, = 0 after taking coh.

Consequence. $H_{dR}^i(\tilde{X}/W)$ is equipped with a Frob

$$\phi: \sigma^* H_{dR}^i(\tilde{X}/W) \rightarrow H_{dR}^i(\tilde{X}/W)$$

$$\sigma: W \rightarrow W$$

Idea. take a category

$$\begin{array}{ccc} X & \supset & U \longrightarrow T \\ & \searrow & \downarrow \quad \downarrow \\ & & S_0 \longrightarrow S \end{array}$$

($\text{Spec } k \rightarrow \text{Spec } W_n$)

U open of X

$U \hookrightarrow T$ some inf. thickening

divided power structure on $U \hookrightarrow T$

$$\exists x \in I, \frac{x^n}{n!} \in \mathcal{O}_T.$$

